

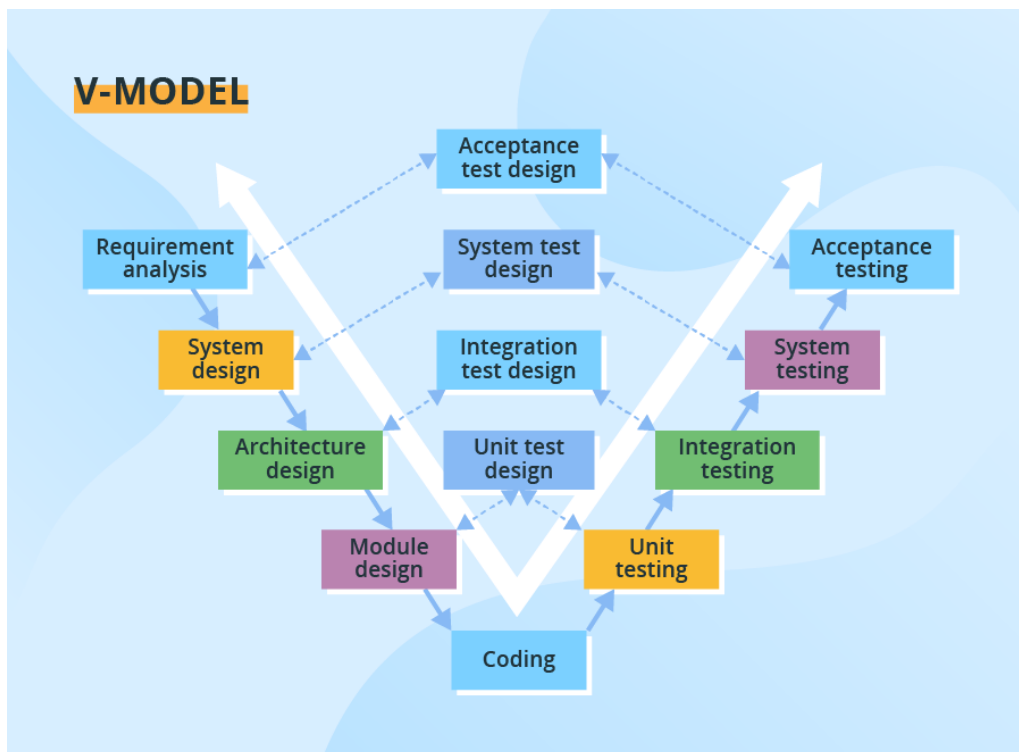
V-Model (Verification and Validation Model)

Overview

The **V-Model**, also known as the *Verification and Validation Model*, is an extension of the Waterfall Model that emphasizes **quality assurance at every stage**. It integrates validation and verification activities alongside development phases, forming a “V” shape that reflects the parallel relationship between development and testing.

- High quality but high cost model.
- Builds upon Waterfall principles with formal testing at each phase.
- Suitable for safety-critical and regulated systems.

V-Model Diagram



Key Idea

The V-Model emphasizes that:

- **Verification** ensures each development phase meets its predefined specifications.
- **Validation** ensures the final product meets user needs.

Each development activity on the left corresponds to a testing activity on the right.

Benefits vs Challenges

Benefits	Challenges
Provides strong risk control through tightly integrated QA.	Increases development costs and time due to extensive QA.
Simplifies compliance with a structured, well-documented process.	Delays project start because of heavy up-front planning.
Reduces the cost of error handling through early defect detection.	Requires extra effort for documentation.
Ensures traceability and transparency across all stages.	Makes mid-project requirement changes costly and difficult to implement.

Best For

- Mission-critical applications where failures are unacceptable, such as:
 - Automotive, aviation, medical, or defense systems.
- Projects in regulated domains that require validation, documentation, and traceability.

Not Ideal For

- Projects with frequently changing or unclear requirements.
- Fast-paced or iterative development environments (e.g., Agile settings).